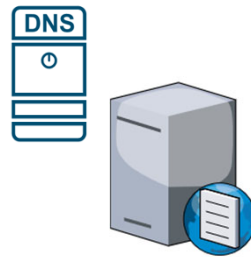


Domain Name System



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

1

Hostnames vs IP addresses

Two identifiers (labels) associated with any device connected to the Internet

- Name: alphanumeric label
 - variable length
 - variable structure (conform to organizational structure)
 - easy to remember
 - difficult to process by network devices (e.g., routers)
- IP address: numeric label
 - fixed length (i.e., 32 or 128 bits)
 - fixed structure (conform to network structure)
 - difficult to remember
 - easy to process by network devices

Example: **poldo.unipv.it** vs **193.204.34.3**

Human vs machine perspectives

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

2

Historical perspective

At the time of ARPANET, all hostname & IP address pairs were stored in a text file called **hosts** (in the directory **/etc**) managed by SRI Network Information Center

- Changes were submitted to SRI by email
- Network administrators periodically downloaded (via **ftp**) new versions of the **hosts** file from SRI
- Network administrators could choose any hostname

The rapid growth of the Internet caused very serious management issues

- *Scalability*: SRI could not handle the load and traffic
- *Ambiguity*: SRI was unable to enforce unique hostnames
- *Inconsistency*: many machines had stale copies of the **hosts** file

Note: The file **hosts** is still used nowadays as a *static* table (e.g., **/etc/hosts** under Linux, **C:\Windows\System32\drivers\etc\hosts** under Windows operating systems)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Example

```
# Copyright (c) 1993-2009 Microsoft Corp.
#
# This is a sample HOSTS file used by Microsoft TCP/IP for Windows.
#
# This file contains the mappings of IP addresses to host names. Each
# entry should be kept on an individual line. The IP address should
# be placed in the first column followed by the corresponding host name.
# The IP address and the host name should be separated by at least one
# space.
#
# Additionally, comments (such as these) may be inserted on individual
# lines or following the machine name denoted by a '#' symbol.
#
# For example:
#
#       102.54.94.97       rhino.acme.com       # source server
#       38.25.63.10      x.acme.com           # x client host
#
# localhost name resolution is handled within DNS itself.
#       127.0.0.1        localhost
#       ::1              localhost
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Simple solution

A fully centralized solution with a single database

- Advantage:
 - Simple to implement
- Disadvantages:
 - Reliability (i.e., single point of failure)
 - Performance/efficiency (i.e., traffic, latency)
 - Management/update
 - Scalability

BOTTLENECK

Problem & solution

The *problem* is how to obtain the *IP address* associated with a given *hostname*

The *solution* consists in the implementation of a specific service dedicated to the *translation* of hostnames into IP addresses

This service is the *Domain Name System*

Domain Name System: basic concepts

Domain Name System (DNS) is a friendly technology used to locate resources on the Internet

First introduced in 1983 by Paul Mockapetris (RFCs 882 and 883 now obsolete), it addresses two main problems:

- How names are mapped into IP addresses
- How applications/services obtain these mappings



Example:

`https://www.google.com`

hostname

From the hostname applications need to derive the corresponding IP address
(*lookup phase*)

Completely transparent

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

What DNS is about

It is defined as “*a general purpose naming service*” [RFCs 1034 “Domain Names – Concepts and Facilities” and 1035 “Domain Names – Implementation and Specification” by Paul Mockapetris published in 1987]

DNS is the core for all networked applications/services (*critical infrastructure*)

It consists of two main components:

1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
2. an application layer *protocol* based on a *client/server model* over *UDP* protocol (on port 53) used to query the database

The basic service provided by DNS is the mapping (i.e., *translation*) between *hostnames* and *IP addresses*

DNS provides additional useful services

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Additional services

Host aliasing

- multiple names map to a single IP address
- one “official” (i.e., *canonical*) name + multiple “nicknames” (i.e., *aliases*)
- used to deploy *virtual hosting*

Mail server aliasing

- a domain name (i.e., alias) maps to one or multiple mail servers (e.g., **@universitadipavia.it**, **@unipv.it**)
- name easy to type/remember by users
- flexibility in the mail service management
- possibility to use the same name for different services

Load distribution across replicated servers (e.g., streaming servers)

- one canonical name maps into multiple IP addresses
- rotation of the addresses provided by DNS
- automatic load balancing

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

DNS

Two main components:

- 1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
2. an application layer *protocol* based on a *client/server model* over *UDP* protocol (on port 53) used to query the database

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

DNS architecture

Core components of the architecture:
distributed database (i.e., mappings of names-addresses) & *Name Servers* (i.e., storage of the mappings)

Hierarchical organization in an inverted tree structure (max tree depth=128)

Each node has a label; sibling nodes must have unique labels

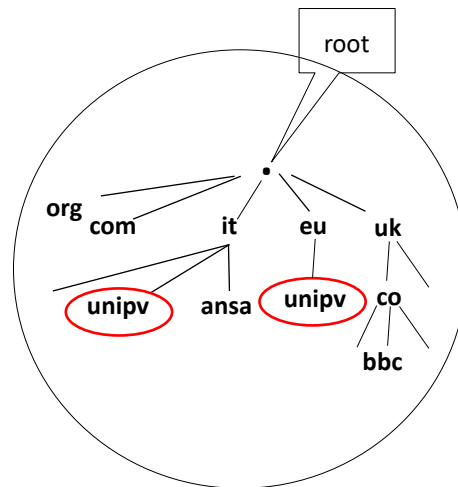
Three main classes of nodes:

- Root
- Top Level Domain (TLD)
- Second Level Domain

A subtree (or a node) denotes a “zone”
(i.e., an administrative space)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure



A domain name is a UNIQUE sequence of labels (separated by a dot) from a node to the root

Root Name Servers

The Root is the entry node of the DNS tree identified by a dot “.” (invisible character)

13 Root Name Servers distributed worldwide and belonging to a reserved domain (i.e., **root-servers.net**) and labelled with a letter from **a** to **m**

The choice of 13 was mainly technical (i.e., answers to DNS queries could fit a single UDP datagram)

Root Name Servers are operated by 12 independent organizations

Root Name Servers are a *very critical piece* of the Internet infrastructure

Many locations/instances of each Root Name Server to ensure resilience and robustness

Total of 1,907 instances around the world (as of 03/2025)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Recap

DNS is a core service for the Internet

It provides translation services and additional services useful for a flexible and efficient management of the technological infrastructure

DNS architecture is characterized by a tree structure: nodes correspond to domains, edges to the relationships between domains

Multiple Name Servers (storing a “subset” of the distributed DNS database) are associated with each node

Three main classes of Name Servers: Root, Top Level Domain, Second Level Domain

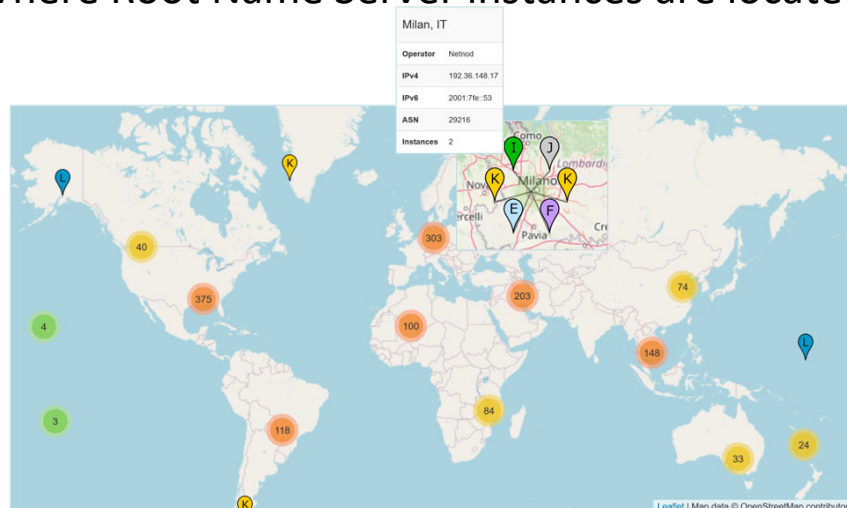
Root Name Servers are the entry point of the DNS tree

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

14

Where Root Name Server instances are located



[Source: <https://www.root-servers.org>]

1,907 instances as March 19, 2025

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

15

List of Root Name Servers

HOSTNAME	IP ADDRESSES	MANAGER
a.root-servers.net	198.41.0.4, 2001:503:ba3e::2:30	VeriSign, Inc.
b.root-servers.net	199.9.14.201, 2001:500:200::b	University of Southern California (ISI)
c.root-servers.net	192.33.4.12, 2001:500:2::c	Cogent Communications
d.root-servers.net	199.7.91.13, 2001:500:2d::d	University of Maryland
e.root-servers.net	192.203.230.10, 2001:500:a8::e	NASA (Ames Research Center)
f.root-servers.net	192.5.5.241, 2001:500:2f::f	Internet Systems Consortium, Inc.
g.root-servers.net	192.112.36.4, 2001:500:12::d0d	US Department of Defense (NIC)
h.root-servers.net	198.97.190.53, 2001:500:1::53	US Army (Research Lab)
i.root-servers.net	192.36.148.17, 2001:7fe::53	Netnod
j.root-servers.net	192.58.128.30, 2001:503:c27::2:30	VeriSign, Inc.
k.root-servers.net	193.0.14.129, 2001:7fd::1	RIPE NCC
l.root-servers.net	199.7.83.42, 2001:500:9f::42	ICANN
m.root-servers.net	202.12.27.33, 2001:dc3::35	WIDE Project

[Source: IANA]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

16

i.root-servers.net Root Name Server

A B C D E F G H I J K L M

OperatorNetnod

[Homepage](#) [Peering Policy](#) [Email](#)

Locations

Sites: 85

[Amaraavati, IN](#) [Amsterdam, NL](#) [Ankara, TR](#) [Ashburn, US](#) [Asunción, PY](#) [Bangkok, TH](#) [Beijing, CN](#) [Belgrade, RS](#) [Berlin, DE](#) [Bratislava, SK](#) [Brussels, BE](#) [Bucharest, RO](#) [Buenos Aires, AR](#) [Calgary, CA](#) [Chicago, US](#) [Cochabamba, BO](#) [Concepción, CL](#) [Denpasar, ID](#) [Doha, QA](#) [Dubai, AE](#) [Florida, US](#) [Frankfurt, DE](#) [Freemont, US](#) [Gothenburg, SE](#) [Guatemala City, GT](#) [Guayaquil, EC](#) [Helsinki, FI](#) [Hong Kong, CN](#) [Jacksonville, US](#) [Jakarta, ID](#) [Johannesburg, ZA](#) [Kigali, RW](#) [Kuala Lumpur, MY](#) [Kyiv, UA](#) [Lima, PE](#) [Ljubljana, SI](#) [London, GB](#) [Luleå, SE](#) [Luxembourg, LU](#) [Malmö, SE](#) [Manama, BH](#) [Manila, PH](#) [Mexico City, MX](#) [Miami, US](#) [Milan, IT](#) [Minneapolis, US](#) [Monterrey, MX](#) [Montevideo, UT](#) [Montreal, CA](#) [New Delhi, IN](#) [Oslo, NO](#) [Palo Alto, US](#) [Panama City, PA](#) [Paramaribo, SR](#) [Paris, FR](#) [Pavlodar, KZ](#) [Perth, AU](#) [Phnom Penh, KH](#) [Porto Alegre, BR](#) [Port Vila, VU](#) [Rabat, MA](#) [Reykjavik, IS](#) [Riga, LV](#) [Riyadh, SA](#) [São Paulo, BR](#) [Semey, KZ](#) [Shenyang, CN](#) [Shiraz, IR](#) [Singapore, SG](#) [Sofia, BG](#) [Stockholm, SE](#) [St Petersburg, RU](#) [Sundsvall, SE](#) [Taipei, TW](#) [Tallinn, EE](#) [Thimphu, BT](#) [Tokyo, JP](#) [Toronto, CA](#) [Ulaanbaatar, MN](#) [Vancouver, CA](#) [Vienna, AT](#) [Vilnius, LT](#) [Wellington, NZ](#) [Yerevan, AM](#) [Yogyakarta, ID](#)

IPv4192.36.148.17

IPv62001:7fe::53

ASN29216

Legend

[IPv6 Enabled Global](#) [IPv4 Only Global](#) [IPv6 Enabled Local](#) [IPv4 Only Local](#)

[Source: <https://www.root-servers.org>]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

17

m.root-servers.net Root Name Server

A B C D E F G H I J K L M

Operator: WIDE Project

Locations: Sites: 23

Bangkok, TH, Brisbane, AU, Dhaka, BD, Guam, GU, Hanoi, VN, Hong Kong, HK, Jakarta, ID, Kaohsiung, TW, Kathmandu, NP, Kolkata, IN, Kuala Lumpur, MY, Lahore, PK, Male, MV, Mumbai, IN, Osaka, JP, Paris, FR, Phnom Penh, KH, San Francisco, US, Sao Paulo, BR, Seoul, KR, Singapore, SG, Tokyo, JP, Ulaanbaatar, MN

IPv4: 202.12.27.33

IPv6: 2001:dc3::35

ASN: 7500

Legend

IPv6 Enabled Global, IPv4 Only Global, IPv6 Enabled Local, IPv4 Only Local

[Source: <https://www.root-servers.org>]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

18

Root Name Servers implementation

Two addresses, i.e., one IPv4 and one IPv6, associated with **all** instances of a given Root Name Server (e.g., 192.36.98.17 and 2001:7fd::1 for Root NS i)

Anycasting scheme used to make instances collectively behave as a single server (e.g., routers choose the “best” instance according to their routing tables) [RFC 1546 “Host Anycasting Service” published in 1993]

Improvements in terms of

- reliability
- scalability
- security

Not every Root Name Server is “anycasted”, i.e., globally replicated

Physical security of Root Name Servers is as important as cyber security

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

19

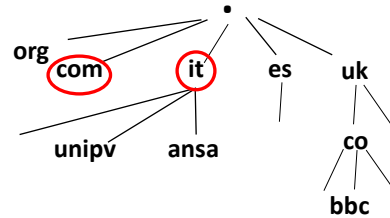
Top Level Domains

Top Level Domain nodes are the children of the Root

These nodes are identified by the right most label of a hostname or domain name

Examples:

`www.unipv.it` `google.com`



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

20

Types of TLDs

In 1998 the Internet Corporation of Assigned Names and Numbers (ICANN) was established to coordinate the Internet naming system

Two types of TLD nodes:

1. **generic (gTLD)** identified by a string of at least three characters (e.g., `info`, `com`, `eco`, `google`, `net`)
2. **country-specific (ccTLD)** identified by a string of two characters according to the ISO standard 3166 "International standard for country codes and codes for their subdivisions" (e.g., `it`, `es`, `ir`, `tr`)

Since 2009 Internationalized Domain Names (IDNs) enable people around the world to use domain names in local languages and scripts (e.g., Arabic, Persian, Chinese)

Domain names start with **xn-** suffix and are encoded with Unicode standard

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

21

gTLD

Two categories of gTLD:

- *sponsored* by organizations/stakeholders (e.g., **edu**, **mil**, **gov**, **museum**)
- *unsponsored*, i.e., operated by policies established by the Internet community (ICANN) (e.g., **com**, **net**, **org**)

In 2011 ICANN launched the “*New gTLD program*” opening the possibility to everybody (mainly private/public organizations) to register (for a fee) a gTLD name

As of March 19, 2025, [1,443 TLDs](#) (gTLD & ccTLD) have been registered

Many companies registered generic TLD names to protect trademarks and avoid *domain name squatting* or *cybersquatting*, that is, an illegal activity of registering gTLDs to make profit from reselling them, selling ads or abusing the domain

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

22

CYBERSQUATTING

Cybersquatting is the practice of third parties registering earlier existing trade marks as domain names, without the authorisation of the trade mark owner, in order to obtain a financial revenue.

Is there any protection mechanism against cybersquatting?

Yes, ICANN launched the [TM Clearinghouse](#) (TMCH) a mechanism which:

Grants priority access to trade mark owners to request domain names.

Notifies the trade mark owner when an identical domain name has been registered.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

23

ccTLD

About 250 ccTLDs operated by organizations (e.g., ISPs, telco companies, universities, research centers) under the “control” of IANA (Internet Assigned Numbers Authority)

ccTLD should be reserved to companies, organizations and individuals based in the country

IANA established the ccTLD **it** in December 1987 and assigned its management to the Italian Research Council – CNR (i.e., **registro.it**)

it is currently operated by IIT-CNR at Pisa and can be used by people/companies based in Italy, EU, Vatican City, San Marino and Switzerland

eu was created in 2006 to promote the European dimension and it can be used by companies, institutions and citizens of EU

Some ccTLD are used as “generic” (e.g., **me**, **am**, **fm**, **tv**)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Usage of ccTLDs

Countries that have cashed in on their top-level domain names

Tuvalu is a tiny island nation of 11,000 people. It’s cashing in thanks to Twitch.

How the tiny Caribbean island of Anguilla has turned the AI boom into a digital gold mine

Alexander Lee
September 23, 2019 at 10:56 a.m. EST

The A.I. Boom Makes Millions for an Unlikely Industry Player: Anguilla

The small Caribbean territory brought in \$32 million last year, more than 10 percent of its G.D.P., from companies registering web addresses that end in .ai.

Google g.co

About g.co

You’ve arrived at this page because you typed or linked to “g.co”, Google’s official URL shortcut just for Google websites.

Whenever you see a short “g.co” link, you can trust that it will always take you to a Google product or service.

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Second level nodes

These nodes are the children of the TLD nodes and refer to the second level of the DNS hierarchy

These nodes are managed by companies, organizations or individuals who *registered* the domain (or by providers offering DNS services)

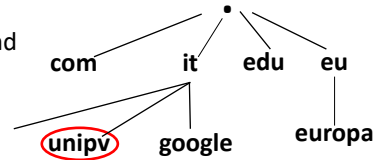
They are “authoritative” for a given zone

Second level nodes might have children (and grandchildren)

Examples:

unipv.it, **bbc**.com, **berkeley**.edu

imperial.ac.uk, **bankofindia**.co.in, **bbc**.co.uk

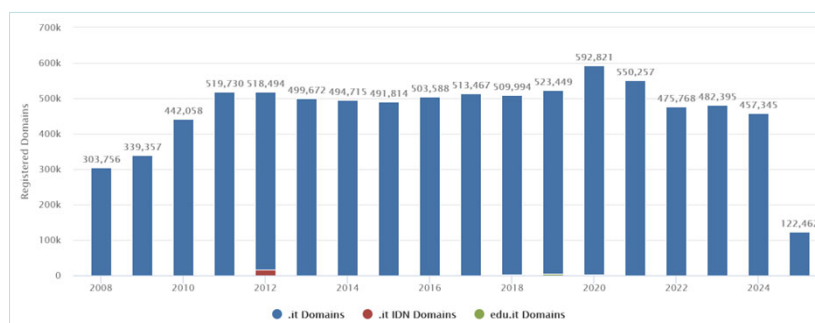


© M. C. Calzarossa 2025

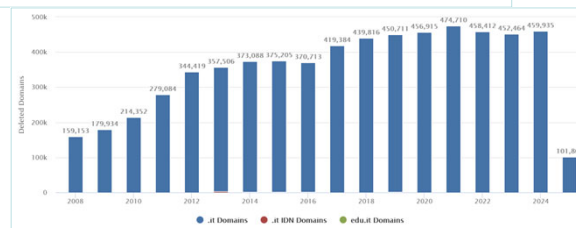
Enterprise Digital Infrastructure

26

Second level domains under it



More than 3.5M domain names registered under **it** since Dec. 23, 1987



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

[Source: <https://stats.nic.it/domain/registration>]

27

Domainsquatting & typosquatting

Microsoft vs. MikeRoweSoft

[Share](#) [Tweet](#)

MikeRoweSoft settles for an Xbox

Canadian student Mike Rowe, who Microsoft put under legal pressure over his Web site's name, has agreed to bury the hatchet out of court for an Xbox and some training.

Cybersquatting: Attackers Mimicking Domains of Major Brands Including Facebook, Apple, Amazon and Netflix to Scam Consumers

Domain Squatting and Brand Hijacking: A Silent Threat to Digital Enterprises

Domain squatting can lead you to malicious websites, and it might be too late to realize what actually happened.

BY WAQAS NOVEMBER 15, 2023 · 6 MINUTE READ

Domain Typosquatting: Online trademark protection in the name of typing errors

NEWSROOM NOVEMBER 14, 2023

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

28

Management

Each node of the hierarchy is identified by a name (label) and operated under the responsibility of a company or an organization (i.e., *authority* of the node)

Name Servers are associated with each node of the DNS hierarchy

Name Servers store the mappings between hostnames and IP addresses (i.e., a subset of all possible mappings, *no single repository*)

Numerous geographically distributed Name Servers (and instances)

Operators of individual nodes must ensure

- 24/7 availability
- Good security levels
- Good performance levels

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

29

Name Server diversity

At least **two** Name Servers per domain/node (which must **not** resolve to the same IP address)

- Network diversity
- Geographical diversity
- Institutional diversity
- Software and hardware diversity

Three Name Servers associated with the domain **unipv.it**

ipv36.unipv.it → 193.204.35.27 [AS137 GARR]

ipv512.unipv.it → 131.175.192.3 [AS198949 RADWARE]

ipv36cloud.unipv.it → 131.175.200.98 [AS198949 RADWARE]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Recap

DNS architecture is characterized by a *hierarchical* structure

Name Servers are associated with the individual nodes of the hierarchy

At least two Name Servers per domain to ensure a reliable and efficient service

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

31

How to register a (2nd level) domain

Registration: adding a node at any level of the DNS tree by purchasing a domain name

Several steps:

1. Choice of the TLD
2. Choice of the Registrar
3. Choice of the name for the new domain
4. Choice of the names of **at least two** Name Servers (i.e., one primary and at least one secondary)

Domain names are assigned on a *first-come first-served* basis and must be **unique**

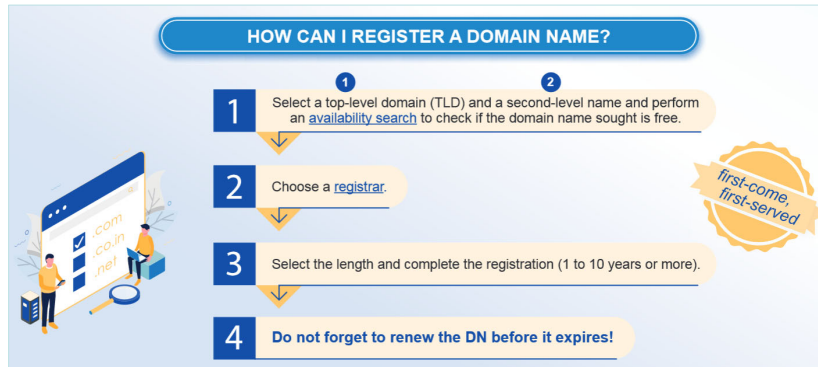
Registrars offer brokerage services to obtain a domain name already in use

Hostnames inside the domain can be freely chosen (but must be unique)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

32



© M. C. Calzarossa 2025

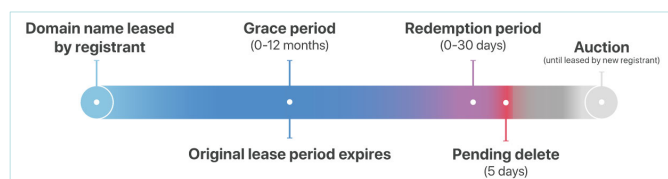
Enterprise Digital Infrastructure

33

Registration

Annual fee associated with each domain name

Renewal is an important step (e.g., autorenewal)



Watch lists of expired domains (e.g., [expired](#)) & domains for sale (e.g., [auctions](#))

Google Argentina's domain name bought by man for £2

36 April 2021

Google Argentina's domain name was bought by a web designer while the site was out of action for two hours in the country last Wednesday.

Hacking law firms with abandoned domain names

How expired internet domains pose a significant cyber risk to the legal profession

Gabor Szathmari [Follow](#)
Aug 21, 2018 · 20 min read

[Twitter](#) [LinkedIn](#) [Facebook](#) [RSS](#)



Expired Domain Allowed Researcher to Hijack Country's TLD

By Eduard Kovacs on January 18, 2021
[Tweet](#) [RSS](#)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

34

DNS database

The *unique* and *authentic* mappings between hostnames and IP addresses are stored in a wide-area *distributed* database

Largest distributed database in the world characterized by:

- Scalability
- Decentralized maintenance
- Robustness
- Global scope (i.e., names have the same meaning everywhere)
- No need of atomicity
- No need of strong consistency

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

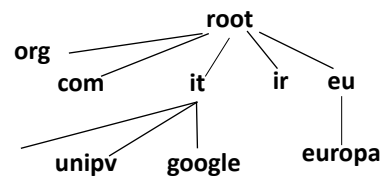
35

Name Servers & database

How mappings between hostnames and IP addresses are distributed across Name Servers

Each Name Server stores only a subset of the entire database

Every node of the hierarchy knows about its children (and its parent)



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

36

Who knows what

Root Name Servers know hostnames and IP addresses of **all** Name Servers associated with **all** TLDs

Each Name Server of a TLD knows hostnames and IP addresses of **all** Name Servers associated with its Second Level domains

Each Second Level Name Server knows hostnames and IP addresses of **all** devices registered under its domain name & hostnames/IP addresses of the Name Servers associated with its TLD

Every Name Server knows hostnames and IP addresses of Root Name Servers (4KB of data)

This hierarchical organization allows every Name Server to obtain the IP address (and hostname) of every device

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

37

Content of DB

Information stored in the database are structured in Resource Records

Different types of RRs with a standard format [RFC 1035 "Domain Names – Implementation and Specification"]:

NAME, **TYPE**, CLASS, **TTL**, RDLENGTH, **RDATA**

NAME: label associated with the record (e.g., FQDN)

TYPE: code specifying the type of record [2 bytes]

CLASS: set to a default value **IN**

TTL: how long a record can be stored in cache (e.g., TTL=0 no caching) [4 bytes]

RDATA & **RDLENGTH**: data describing the resource (as a function of the record type) and length of the data field [2 bytes]

Comments: starting with ";" [optional]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

38

Mappings

The RR types have to cope with the different mappings between hostnames and IP addresses supported by DNS:

- One-to-one

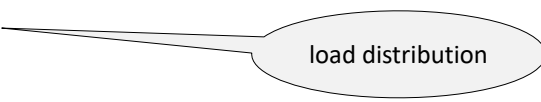
- Many-to-one

- One-to-many

- *One-to-zero*



virtual hosting



load distribution

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

39

RR types

RR types are subdivided into four categories:

- Basic
- Zone
- Security
- Optional

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

40

RR types: basic

Type **A** (Address) or **AAAA**: specifies the IP address corresponding to a given hostname [*forward mapping*]

Example: NAME: FQDN **eva.unipv.it**
 RDATA: IP address **193.204.34.2** [*one-to-one mapping*]

In the one-to-many mappings, all IP addresses are listed in the data field

Type **MX** (Mail eXchanger): specifies the mail server(s) associated with a domain

Example: NAME: domain name **gmail.com**
 RDATA : name of the mail server(s)
 alt2.gmail-smtp-in.1.google.com
 alt3.gmail-smtp-in.1.google.com
 gmail-smtp-in.1.google.com
 alt1.gmail-smtp-in.1.google.com
 alt4.gmail-smtp-in.1.google.com

© M. C. Calzarossa 2025

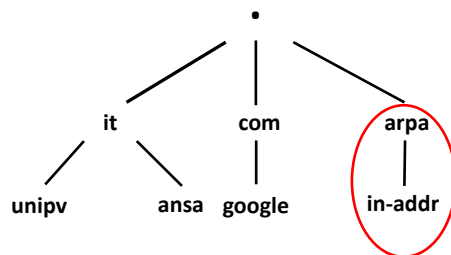
Enterprise Digital Infrastructure

41

RR types: basic (cont'd)

Type **PTR** (Pointer): maps an IP address to a hostname [*reverse mapping*]

Example: NAME: reverseIPAddress.**in-addr.arpa**
 2.34.204.193.in-addr.arpa
 RDATA: FQDN
 eva.unipv.it



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

42

RR types: zone

Type **NS** (Name Server): defines the names of the Authoritative Name Servers of a zone/domain [*infrastructure record*]

Example: NAME: domain name **unipv.it**
 RDATA: names of the Name Servers
 ipv36.unipv.it
 ipv36cloud.unipv.it
 ipv512.unipv.it

Type **SOA** (Start Of Authority): *mandatory* record used to specify the core characteristics and attributes of a zone [*first record of a domain zone*]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

43

SOA

origin: name of the **primary Name Server** of the domain

mail addr: email address of the domain administrator

serial: serial number (e.g., date of last change & version number)

refresh: time [s] indicating the interval before the zone should be refreshed

retry: time [s] indicating the interval that should elapse before a failed refresh should be retried

expire: time [s] indicating the upper limit of the interval that can elapse before the zone is no longer authoritative

TTL: minimum TTL for the RRs of the zone

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

44

SOA Records: examples

```
root-servers.net
primary name server = a.root-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 2025031300
refresh = 14400 (4 hours)
retry = 7200 (2 hours)
expire = 1209600 (14 days)
default TTL = 3600000 (41 days 16 hours)

it
primary name server = dns.nic.it
responsible mail addr = hostmaster.nic.it
serial = 2025032309
refresh = 10800 (3 hours)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 3600 (1 hour)

com
primary name server = a.gtld-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 1742727127
refresh = 1800 (30 mins)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 900 (15 mins)

co.nz
primary name server = loopback.dns.net.nz
responsible mail addr = soa.nzrs.net.nz
serial = 2503231046
refresh = 900 (15 mins)
retry = 300 (5 mins)
expire = 604800 (7 days)
default TTL = 3600 (1 hour)
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

45

Second Level Domain SOA Records: examples

```
unipv.it
primary name server = ipv36.unipv.it
responsible mail addr = root.ipv36.unipv.it
serial = 2025031902
refresh = 12000 (3 hours 20 mins)
retry = 1200 (20 mins)
expire = 1209600 (14 days)
default TTL = 300 (5 mins)

polimi.it
primary name server = ns.polimi.it
responsible mail addr = root.ns.polimi.it
serial = 202006810
refresh = 21600 (6 hours)
retry = 3600 (1 hour)
expire = 604800 (7 days)
default TTL = 86400 (1 day)

amazon.com
primary name server = dns-external-master.amazon.com
responsible mail addr = hostmaster.amazon.com
serial = 2010193634
refresh = 180 (3 mins)
retry = 60 (1 min)
expire = 604800 (7 days)
default TTL = 900 (15 mins)

twitch.tv
primary name server = ns-219.awsdns-27.com
responsible mail addr = awsdns-hostmaster.amazon.com
serial = 1
refresh = 300 (5 mins)
retry = 150 (2 mins 30 secs)
expire = 604800 (7 days)
default TTL = 60 (1 min)

google.it
primary name server = ns1.google.com
responsible mail addr = dns-admin.google.com
serial = 739443398
refresh = 900 (15 mins)
retry = 900 (15 mins)
expire = 1800 (30 mins)
default TTL = 60 (1 min)
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

46

Recap

DNS database is distributed worldwide across multiple Name Servers (with multiple replicas)

Each Name Server stores a specific portion of the database

Information organized in Resource Records with a standard format

Different types of records to cope with the various services provided by DNS

IP addresses stored *only* in records of type A or AAAA

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

47

RR types: security

These record types refer to the security extensions that provide source authentication for the DNS [RFC 4034 “Resource Records for the DNS Security Extensions” published in 2005]

Type **DNSKEY** : specifies the public key associated with a domain

Example: NAME: domain name
 RDATA: public key

Type **RRSIG** : specifies the digital signature for a set of Resource Records

Type **NSEC (Next Secure)**: specifies the types of Resource Records associated with a domain and the next authoritative name after the domain (used for *authenticated denial of existence*)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

48

RR types: optional

These record types refer to *optional* features supported by DNS

Type **CNAME** (Canonical Name): specifies the canonical name corresponding to an alias (e.g., CDN)

Example:	NAME: alias	<code>www.mit.edu</code>
	RDATA: canonical name	<code>www.mit.edu.edgekey.net</code>
	NAME: alias	<code>www.mit.edu.edgekey.net</code>
	RDATA: canonical name	<code>e9566.dscb.akamaiedge.net</code>

Type **TEXT** (Text): used to associate text with a domain name to be used for different objectives

Example:	NAME: domain name	<code>unipv.it</code>
	RDATA: textual content	
	<code>"MS=421AA9E190F698E46727F29DA1D625BF605AEF88"</code>	
	<code>"facebook-domain-verification=8hfo9hldtvmuhvv3ptr8g7884oq9e"</code>	
	<code>"google-site-verification=ZY0Orml0atjmdx-kUyATJatNbSGVCzxm2xhAWVH0lc"</code>	
	<code>"atlassian-domain-verification="</code>	

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

49

Other RR types

The service record (**SRV**) is used to discover the services offered by a domain [RFC 2782 "A DNS RR for specifying the location of services (DNS SRV)" published in 2000]

A service is identified by a symbolic name, the transport protocol used and the domain name (i.e., **_Service._Proto.Name**)

SVCB (Service Binding) and **HTTPS** RRs aim at facilitating the lookup of information needed to connect to network services ahead of the connection [RFC9461 "Service binding and parameters for DNS Servers (SVCB and HTTPS Resource Records)" November 2023]

Benefits for performance and privacy (RTTs and unencrypted messages)

`abc.com 300 IN HTTPS 1 cdn.abc.net alpn="h3","h2"`

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

50

Root hint and zone files

The root hints are a set of RRs used to locate Name Servers that are *authoritative* for the root of the DNS tree

Root hints

Each Name Server stores its own *zone file*

A zone file is a text file that contains the RRs referring to the zone associated with the Name Server

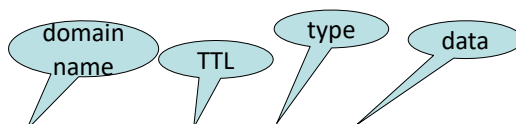
Root zone

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

51

RRs associated with **it.**



organisation: IIT - CNR
address: Via Moruzzi, 1
address: Pisa I-56124
address: Italy

created: 1987-12-23
changed: 2025-01-28
source: IANA

```
it. 172800 IN NS a.dns.it.
it. 172800 IN NS m.dns.it.
it. 172800 IN NS r.dns.it.
it. 172800 IN NS v.dns.it.
it. 172800 IN NS dns.nic.it.
it. 172800 IN NS nameserver.cnr.it.
it. 86400 IN DS 41901 10 2 47f7f7ba21e48591f6172eed13e35866893ad9f2880fc9bada64f68ce28ebb90
it. 86400 IN RRSIG DS 8 1 86400 20250410050000 20250328040000 26470 .
U6CbfjT2zhwFkQ0t142+w90aYHn0paS1TCuOHCjgqtkwPKOp4j3K5vOCxMBtYFHE2Q+N1IefVx344CylvaY9mAMVxmv4aRLpFxJ37CzaQkTc1xHH1U11T9Hst1+yRHM3gmDRLf58XVjrs5l
EE5jtQsDerA0s4Mo4Sfr9012oiBs0KyGVuABZp1miTjadbH97AE1xCRVpS59Y5R+Z/XSGVxemb/e6gP517KPyok/qZ7c0zU311YkIWbH5bDu2oDz+Mykyc+
+eEujiTU+RpytWl19EPxs8cddlP1ac3QQCmRvYFEBX14CexgoNduHuFpaS21Wna6054kmw==
it. 86400 IN NSEC itau. NS DS RRSIG NSEC
it. 86400 IN RRSIG NSEC 8 1 86400 20250410050000 20250328040000 26470 . ZQjmJq5vyp8F0hu7DC8Cta/
Tc8Yiak3mcXCgBgfriuelSnR5EuQ02HQIhdS95gs87gmfo4MvrIa0z5rDykb73Fs15vcrDA9J6WlkJGAgDafPYCYNPFaUoagRUXIZISmv0DEd+G+MhV9bJBVRtON4e/bpQhRzv4pnq896q
QxX3nk2X3fN1HFK24mAcDk801S8p2UTAjz2e2TgR5anJXG6jri4+2dIN8pqC0o1a+iRq98jevXs60L2TQXmf1BAMp7QIbgBIsbHaxkVc1B1S11yIcflfbJNLraJfr/mzco/nI8CDF3zpwLIC
nameserver.cnr.it. 172800 IN A 194.119.192.34
nameserver.cnr.it. 172800 IN AAAA 2a00:1620:c0:220:194:119:192:34
a.dns.it. 172800 IN A 194.0.16.215
a.dns.it. 172800 IN AAAA 2001:678:12:0:194:0:16:215
m.dns.it. 172800 IN A 217.29.76.4
m.dns.it. 172800 IN AAAA 2001:1ac0:0:200:0:a5d1:6004:2
r.dns.it. 172800 IN A 193.206.141.46
r.dns.it. 172800 IN AAAA 2001:760:ffff:ffff:0:0:0:ca
v.dns.it. 172800 IN A 194.0.25.44
v.dns.it. 172800 IN AAAA 2001:678:20:0:0:0:0:44
osiris.nameex.it 172800 IN A 193.201.40.6
osiris.nameex.it 172800 IN AAAA 2a0f:80:1:0:0:0:0:6
seth.nameex.it. 172800 IN A 193.201.40.7
seth.nameex.it. 172800 IN AAAA 2a0f:80:1:0:0:0:0:7
dns.nic.it. 172800 IN A 192.12.192.5
dns.nic.it. 172800 IN AAAA 2a00:d40:1:1:0:0:0:5
itau. 172800 IN NS a.nic.itau.
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

52

Examples of RRs for google.

google.	172800	IN	NS	ns-tld1.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld2.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld3.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld4.charlestonroadregistry.com.
google.	172800	IN	NS	ns-tld5.charlestonroadregistry.com.
google.	86400	IN	DS	6125 8 2 88f88780231071535788AD3800E9543500474E5C30C29698B40A3DB23ED9DA9F
google.	86400	IN	RRSIG	DS 8 1 86400 20230410050000 20230328040000 951 .
EKfUqg/AGEnl0tj4LSGGrae8Iggst08C6sp+5ZkYLXWGHNCaFoFTPDL58FHCc6pT3CH21J/nY5wKDT9dvB87xKpGkF5F+9pN+k/so3X1R/iacDd3wxHDBpc				
/GKa7nFlKocK+RQk5P3VDWqZMfzMQh4nxWAcGq5C3wFN+hLRrQQGYUMYHOSYkamcN/123TxCZ1Yf513wFV2				
/iWhWuHfri7bcMGBruPOFwemBgZz87mqubv26nVvZc0p2055w3SHgrTUNjrRC4kXo1rj01uHj0e77wAZHzvg0/Fv1yZMVRfvTFMFsFgnQ5g0+ic0+Lc09xNudpw7gx+QzCLRsCdG				
google.	86400	IN	NSEC	gop. NS DS RRSIG NSEC
google.	86400	IN	RRSIG	NSEC 8 1 86400 20230410050000 20230328040000 951 .
JRDvxRa/biNMiJLetv0P6qNA+GvafG3jws5cVT7ZMJmp0sP+AKzbGmwOyVzmngJESDzh				
/AjYRkpQ19IF39Cd1QXLCU5N3bxALu65YtEO+m8QcwD3w5gZcJvgXgF3sMBZd6oV18Rbcgq10D57PUeto54jGo3hbwrzi+fk2Kgle10gfGC3				
/zQroGhXrvNDrTvPI70FenQ21MQH1QIH1eC				
/w4UBDYEPrsYAdMLzkGhdh6x0MK1B3CoXSU7YTEu5+1jDRhsuemJLNnleUXoFvmMLFF6LK9bt4AAtwIAnIhYf5ak01hmTA3mv2onPMTXZ78bYvrM30EKKYImM28GBas6R==				

Registrant:

Name: Charleston Road Registry Inc.

Organization: Charleston Road Registry Inc.

Mailing Address: 1600 Amphitheatre Parkway | Mountain View CA 94043 United States of America (the)

Administrative:

Name: TLD Admin

Organization: Google Inc.

Email: iana-contact@google.com

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

53

Database updates

At least two Name Servers associated with a zone:

- Primary (i.e., master) Name Server
- Secondary (i.e., slave) Name Server(s)

RRs are *manually* entered into a zone file and loaded into the primary (authoritative) Name Server for the zone

Each secondary NS could poll the primary NS or it might *automatically* receive notifications from the primary NS (e.g., **NOTIFY** option)

Changes to the master copy of the database are propagated to the secondary NSs according to timings (e.g., *refresh* parameter) set by the domain administrator in the SOA RR (*loose coherency*)

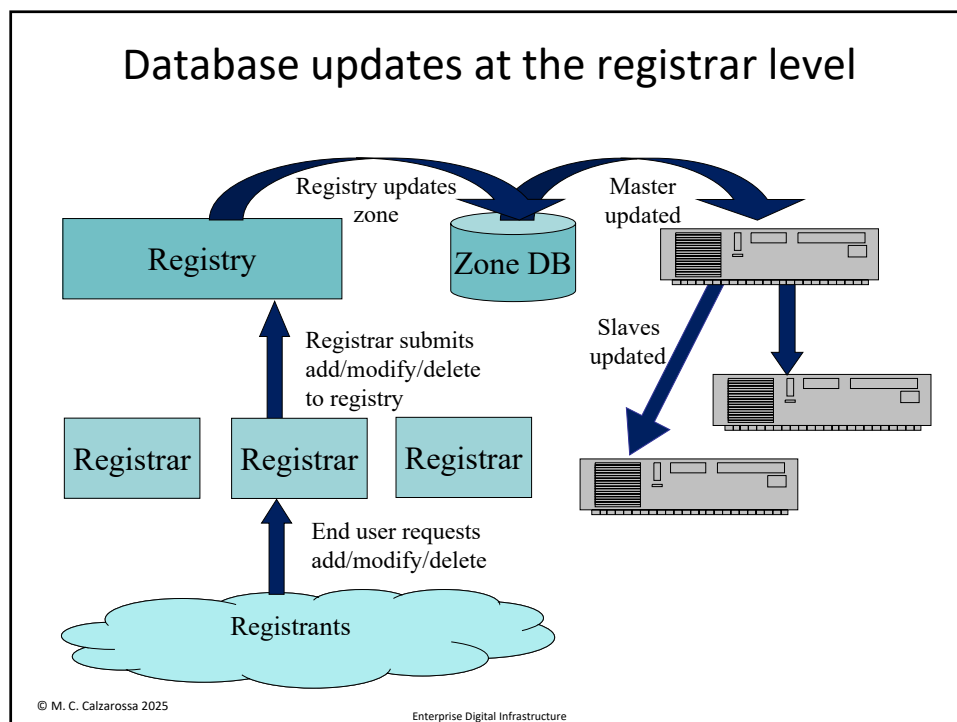
A serial number is associated with each version of a zone file

This number is modified whenever an administrator changes anything

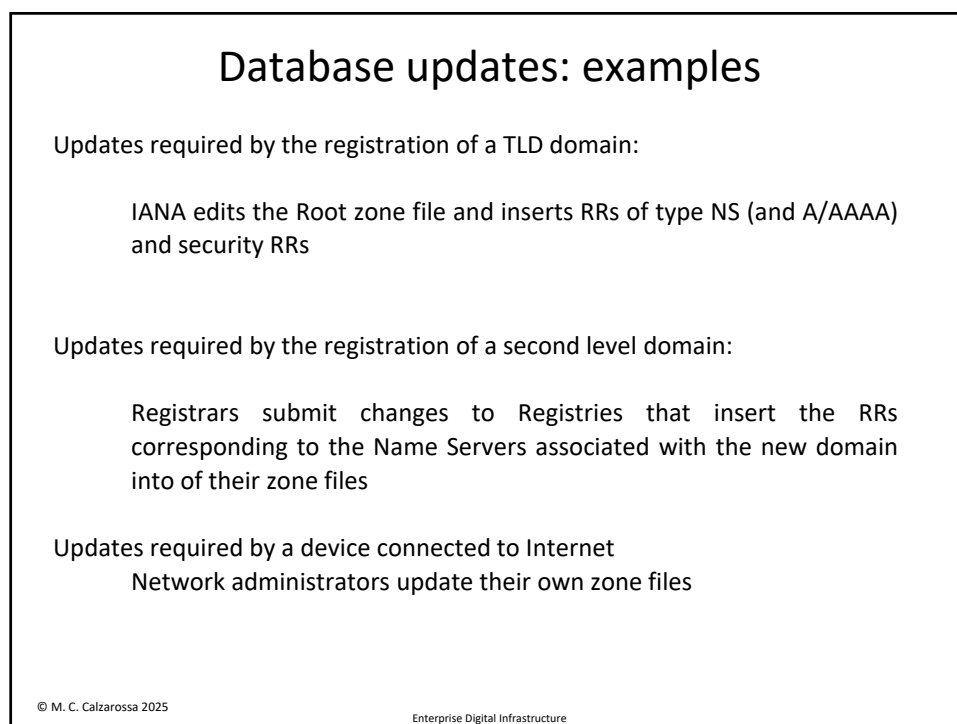
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

54



55



56

DNS setup: user perspective

How do we associate a Name Server with a device?

It depends on the policy adopted by the ISP on the usage of IP addresses

1. Static IP addresses
2. Dynamic IP addresses assigned via DHCP

In case of dynamic IP addresses, DHCP *automatically* assigns the Name Server(s) but we can also set them *manually*

In case of static IP addresses, it is necessary to know the IP address(es) of the Name Server(s) of the domain or of “open” Name Server(s) (e.g., `/etc/systemd/resolved.conf` under Linux)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

57

DNS

Two main components:

- ✓ 1. a *distributed database* implemented on many *geographically distributed servers* characterized by a *hierarchical organization*
- 2. an application layer *protocol* based on a client/server model over the UDP protocol (on port 53) used to query the database (i.e., to provide the translation service)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

58

DNS protocol overview

DNS is a *lightweight application layer protocol* used to query the database

Client/server model

Stub resolver part of the operating system: client component running on user devices

Full resolver running on Name Servers: both client and server components

Two types of messages exchanged by clients and servers

- query messages sent by clients
- response messages sent by servers

UDP protocol for efficiency (reliability not ensured) & port 53

Implementations based on TCP mainly used for maintenance/update purposes or to ensure security/privacy

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Recap

Different types of Resource Records to cope with the various services provided by DNS and to implement additional services

Any change within domains requires updates of the DNS database

DNS protocol at the application layer used to query the database

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

61

DNS protocol: example

`ssh eva.unipv.it`

hostname used by stub resolver

lookup phase

The stub resolver

1. converts into DNS queries the requests received by applications for the translation of the hostname into the corresponding IP address
2. sends queries to a full resolver
3. receives the results and returns them to the application

Two possible outcomes of a query:

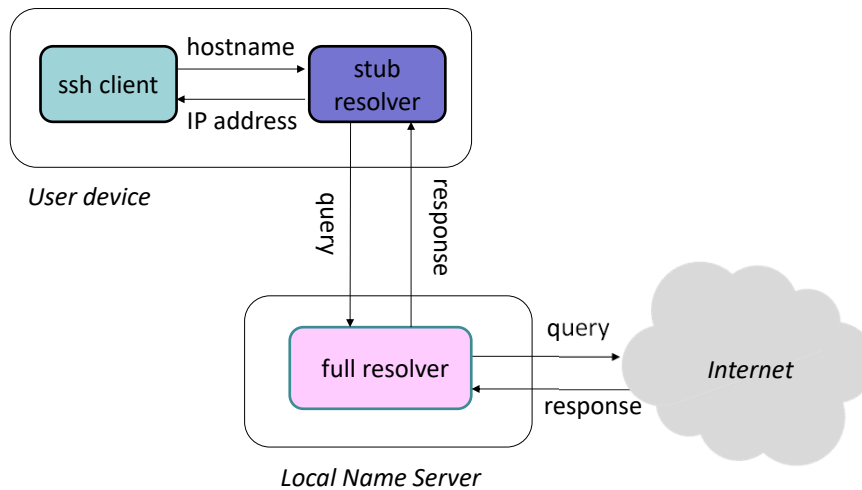
- *success*: the answer provides the IP address corresponding to the name specified by the query
- *failure*: the answer contains an error (e.g., **`NXDOMAIN`**)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

62

Which Name Server(s)



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

63

Iterated vs recursive queries

An *iterated* query is centered around the local Name Server that is responsible to send queries and collect answers

Each NS can provide

- a *complete answer*
- a *partial answer*
- an *error* (due to a permanent or temporary failure)

A *recursive* query puts the responsibility of the answer on the specific Name Server being contacted

Each NS can provide

- a *complete answer*
- an *error* (due to a permanent or temporary failure)

The queries between clients and local Name Servers are **always** recursive, while Root and TLD Name Servers support iterated queries **only**

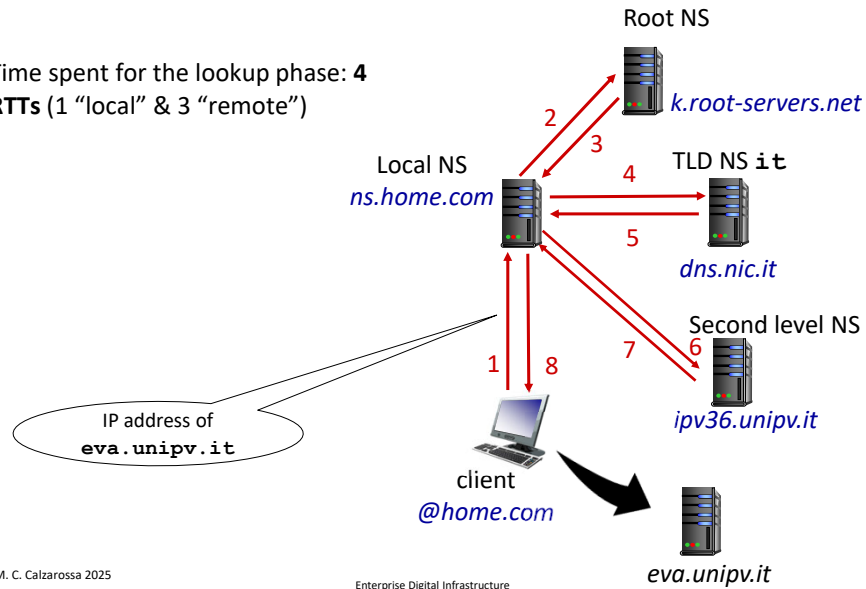
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

64

Example of iterated query

Time spent for the lookup phase: **4**
RTTs (1 "local" & 3 "remote")

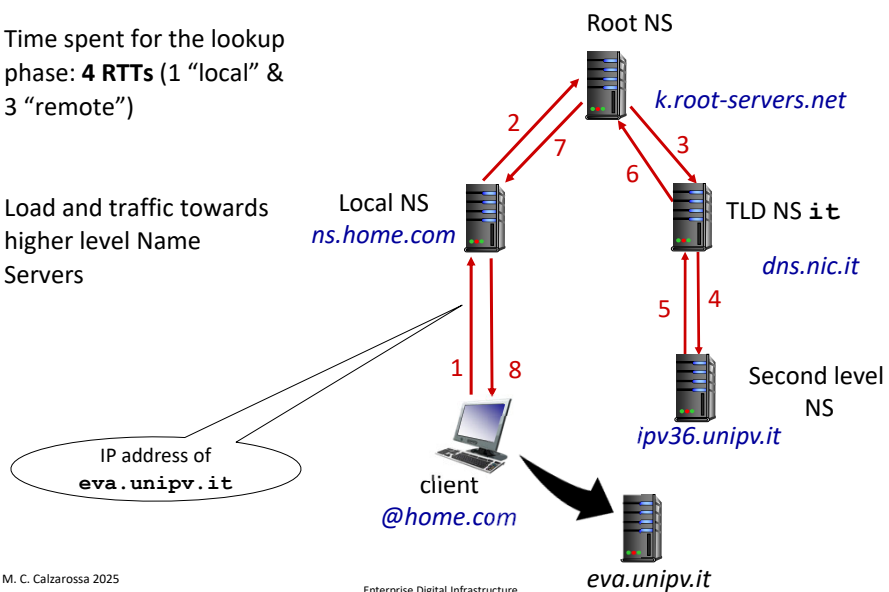


65

Example of recursive query

Time spent for the lookup phase: **4** RTTs (1 "local" & 3 "remote")

Load and traffic towards higher level Name Servers



66

DNS queries/answers

Stub resolvers communicate with *recursive* resolvers running on Local Name Servers (first hop of a query)

Full resolvers running on any Name Server communicate between each other typically using iterated queries

Stub resolvers: clients only

Full resolvers: clients or servers

Answers provided by the Name Servers responsible of the zones/domains are *authoritative*

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Recap

DNS queries involve multiple Name Servers

Iterated and recursive queries

Local Name Server associated with each device (first hop of each query)

Stub resolvers (clients only) communicate with recursive resolvers running on Local Name Servers

Full resolvers (clients or servers) running on the various Name Servers communicate between each other commonly using iterated queries

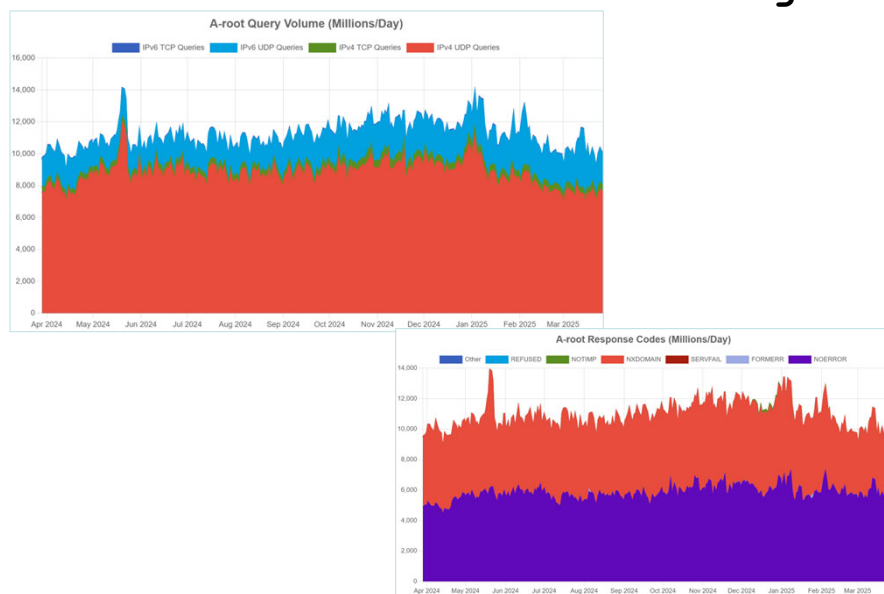
“Official”, i.e., *authoritative*, answers are only those provided by the Name Servers responsible of the zones/domains

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

68

Traffic at **a.root-servers.org**



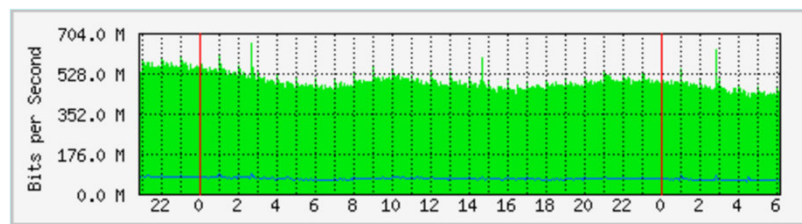
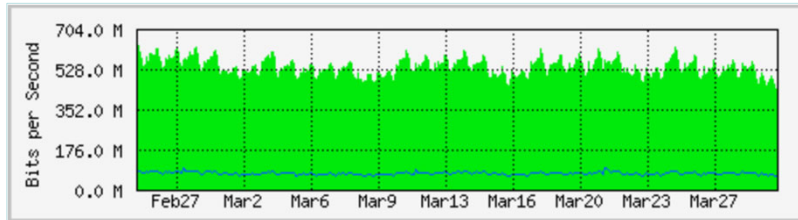
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

[Source: <https://a.root-servers.org/metrics>]

69

Traffic at m.root-servers.org



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

[Source: <https://m.root-servers.org/>]

70

Root Name Servers Performance

Europe	Raw Performance	Uptime	Quality	Last 30 days								
DNS name	Query Speed	0	20	40	60	80	100	120	140	160	180	200
1 e.root-servers.net	6.74 ms											
2 f.root-servers.net	6.95 ms											
3 j.root-servers.net	13.24 ms											
4 d.root-servers.net	15.75 ms											
5 c.root-servers.net	16.77 ms											
6 h.root-servers.net	17.08 ms											
7 a.root-servers.net	17.53 ms											
8 k.root-servers.net	18.35 ms											
9 l.root-servers.net	35.69 ms											
10 i.root-servers.net	35.76 ms											
11 g.root-servers.net	39.22 ms											
12 b.root-servers.net	40.33 ms											
13 m.root-servers.net	56.92 ms											

Europe	Uptime
DNS name	Uptime
1 d.root-servers.net	99.96%
2 e.root-servers.net	99.96%
3 f.root-servers.net	99.96%
4 a.root-servers.net	99.95%
5 g.root-servers.net	99.95%
6 l.root-servers.net	99.95%
7 m.root-servers.net	99.95%
8 h.root-servers.net	99.94%
9 k.root-servers.net	99.93%
10 b.root-servers.net	99.89%
11 c.root-servers.net	98.88%
12 j.root-servers.net	98.83%
13 i.root-servers.net	94.72%

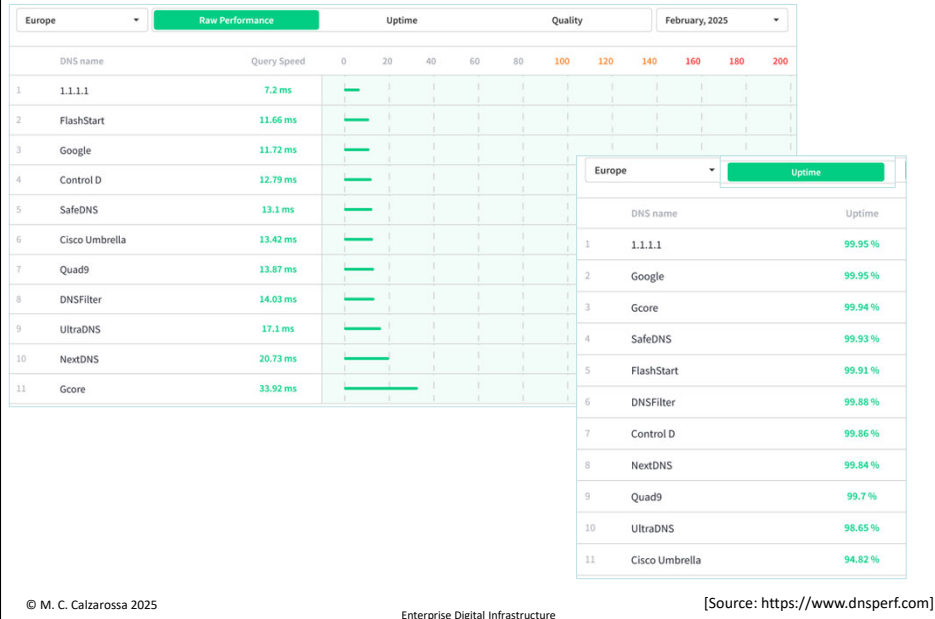
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

[Source: <https://www.dnsperf.com>]

71

Open DNS Performance



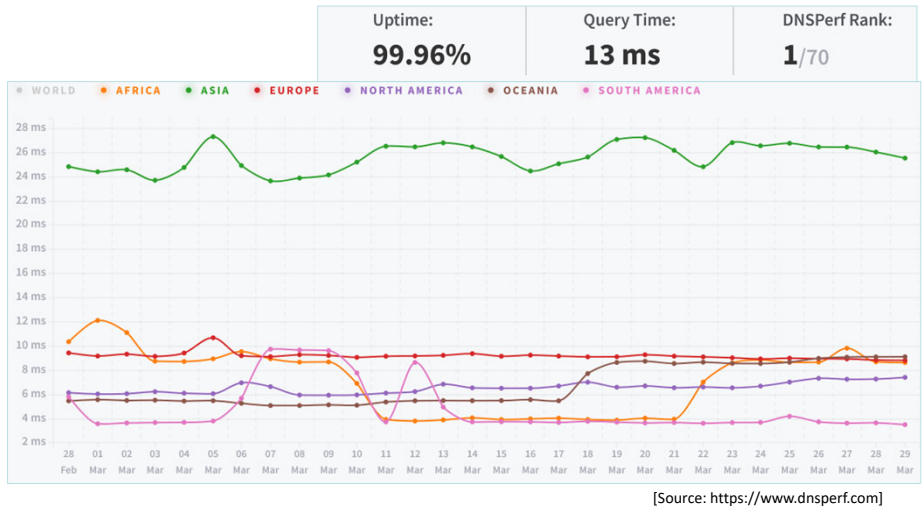
72

Authoritative DNS Performance



73

DNS providers: No-IP

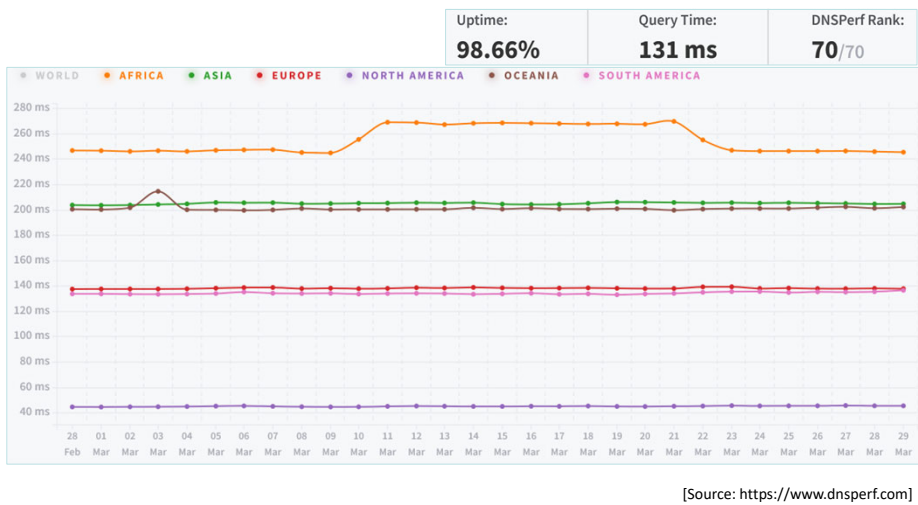


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

74

DNS providers: ZoneEdit



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

75

DNS Messages

A protocol has to specify the *types* of messages exchanged in the interactions between clients and servers as well as the *format* of these messages

Two types of DNS messages

1. Query messages
2. Response messages

Both types of message share the same format

Messages are characterized by variable length

Information are subdivided into *five* sections

One section, i.e., header section, is common to both types of messages

© M. C. Calzarossa 2025

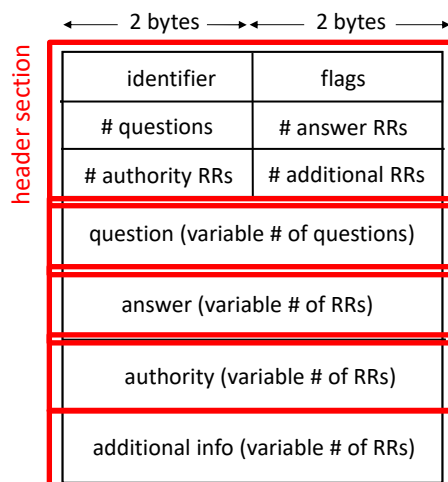
Enterprise Digital Infrastructure

76

DNS message format

Header section (12 bytes) includes:

1. ID used to match query and response
2. Flags
 - query or response
 - recursion desired
 - recursion available
 - authoritative answer
 -
3. Number of questions and number of RRs in the answer/authority/additional sections



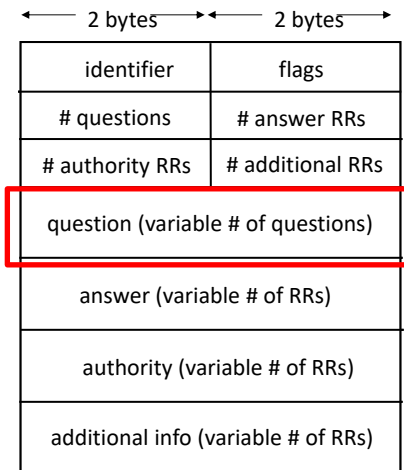
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

77

DNS message format (cont'd)

Question section: contains the query
qname, qclass, qtype



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

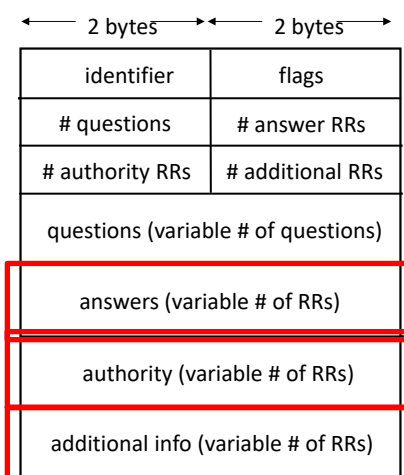
78

DNS message format (cont'd)

Answer section: contains RR(s) in response to the query

Authority section: contains RRs of the Authoritative Name Server of the domain

Additional info section: contains RRs that might be useful (even though not asked in the question)



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

79

Example

```
;; QUESTION SECTION:
;mit.edu.                IN      NS

;; ANSWER SECTION:
mit.edu.                 1800    IN      NS      eur5.akam.net.
mit.edu.                 1800    IN      NS      usw2.akam.net.
mit.edu.                 1800    IN      NS      ns1-37.akam.net.
mit.edu.                 1800    IN      NS      use5.akam.net.
mit.edu.                 1800    IN      NS      ns1-173.akam.net.
mit.edu.                 1800    IN      NS      asia2.akam.net.
mit.edu.                 1800    IN      NS      asia1.akam.net.
mit.edu.                 1800    IN      NS      use2.akam.net.

;; ADDITIONAL SECTION:
ns1-37.akam.net.         3541    IN      A      193.108.91.37
usw2.akam.net.           86917   IN      A      184.26.161.64
ns1-173.akam.net.        7740    IN      A      193.108.91.173
asia2.akam.net.          86953   IN      A      95.101.36.64
eur5.akam.net.           86880   IN      A      23.74.25.64
use2.akam.net.           86836   IN      A      96.7.49.64
asia1.akam.net.          55754   IN      A      95.100.175.64
use5.akam.net.           87346   IN      A      2.16.40.64
use5.akam.net.           106398  IN      AAAA   2600:1403:a::40

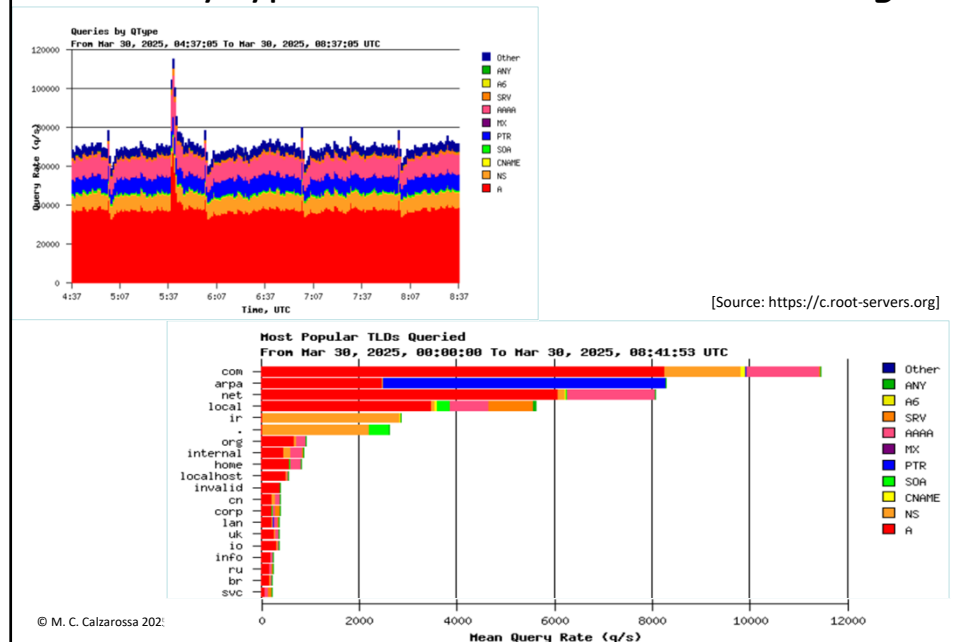
;; Query time: 44 msec
;; SERVER: 192.168.1.1#53(192.168.1.1)
;; WHEN: Sun Mar 30 11:32:29 CEST 2025
;; MSG SIZE rcvd: 387
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

80

Query types at c.root-servers.org



81

Caching

The main objective of caching is to reduce the *time* spent in the lookup phase (i.e., speedup repeated queries) **and** the *load* of the Name Servers

Caches are associated with full resolvers and also with stub resolvers

A DNS cache stores **all** RRs received as answers to queries (even negative answers) for possible reuse by the same query or by different queries (e.g., info about Name Servers)

Why caching is effective:

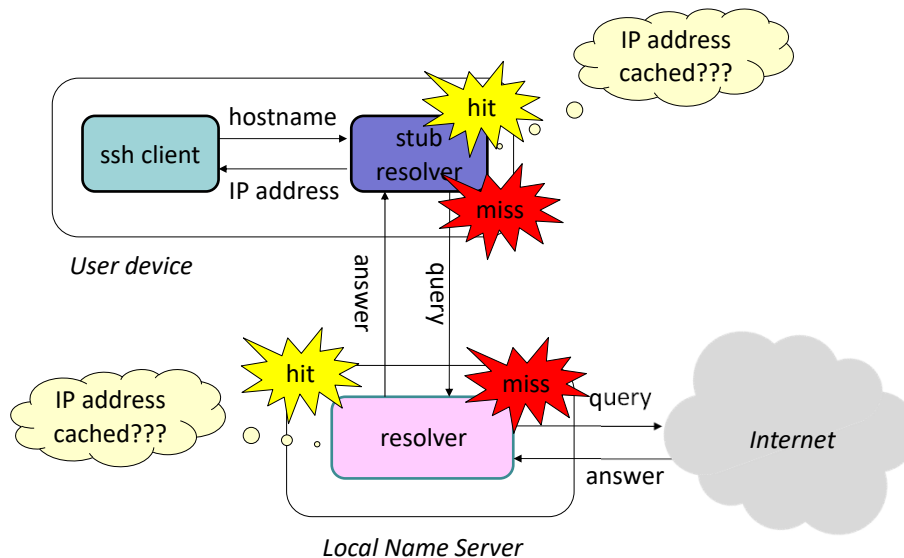
- many cache hits (and few cache misses)
- very few changes to TLD Name Servers
- popular sites/services are visited very often (i.e., a local NS (and even a sub resolver) often stores these IP addresses in its cache)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

82

Cache hits & misses



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

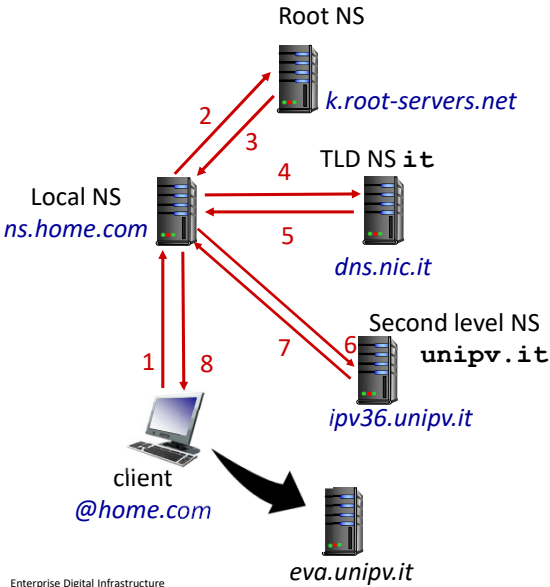
83

Example revisited

Client connected to the Wi-Fi
@home.com opens a terminal
to establish a secure connection
to a remote machine:
ssh eva.unipv.it

The stub resolver queries the
Local Name Server
eva.unipv.it IN A

The resolver of the Local Name
Server queries the resolvers of
the Name Servers of the
hierarchy (the query does not
change)
eva.unipv.it IN A



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

84

Content of local NS cache: first query

The diagram shows the Root NS and Local NS. The Root NS has a red arrow (2) pointing to the Local NS, and the Local NS has a red arrow (3) pointing to the Root NS. The IP address *193.0.14.129* is shown next to the Root NS.

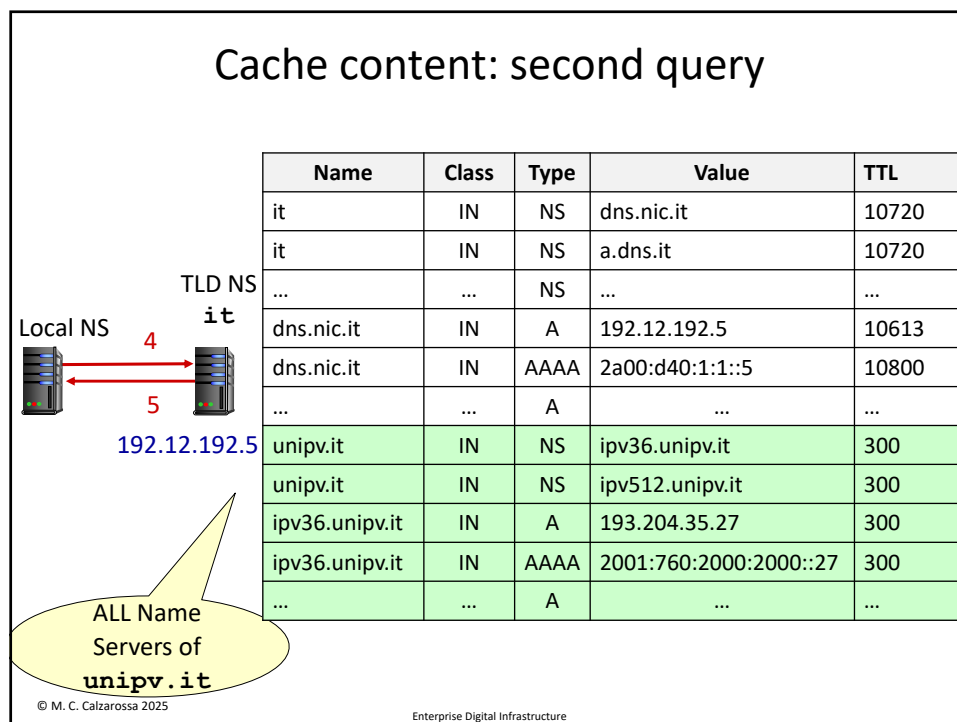
Name	Class	Type	Value	TTL
dns.nic.it	IN	A	192.12.192.5	10613
dns.nic.it	IN	AAAA	2a00:d40:1:1::5	10800
...	...	...	...	...
it	IN	NS	dns.nic.it	10720
it	IN	NS	a.dns.it	10720
...	...	..	...	...

ALL Name Servers of *it*

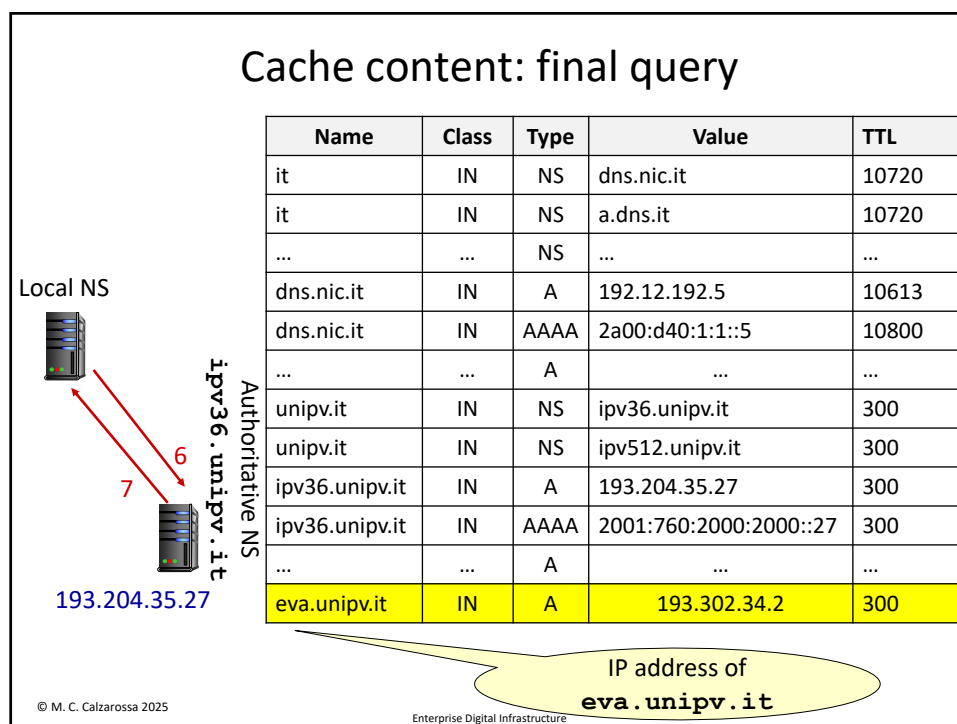
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

85



86



87

Recap

Two types of messages exchanged between resolvers: query messages sent by clients and response messages sent by servers

Caches associated to stub resolvers and full resolvers to speed up queries and reduce the load of Name Servers

Efficiency measured in terms of hit and miss rates

Caches store all Resource Records included in the response messages

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

88

Cache management

Caching is another source of *loose coherence* of the database: not every resolver is necessarily seeing the same content

As soon as cached RRs *expire*, they are removed from the cache

TTL measures the lifetime in the cache of each RR (i.e., how long it can be stored in the cache before requiring a *refresh*)

The default TTL for the Resource Records of a zone is stored in the SOA RR

The value of TTL is expressed in seconds

Choice of the TTL is *critical* for the efficiency of the lookup phase

Basic rules [RFC 1912 “Common DNS Operational and Configuration Errors” published in 1996]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

89

Choice of the TTL

“Small” TTL: RRs stored in the cache will expire quickly

- + quick propagation of the updates
- delays in the lookups

“Big” TTL: RRs stored in the cache will stay there for long time

- + faster lookups
- cache poisoning

In general, “big” TTLs for Root and TLD Name Servers, “small” TTLs for Second level domain Name Servers

The choice of the “best” TTL for a domain is up to the domain administrator

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

90

TTL: examples

```
root-servers.net
primary name server = a.root-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 2025031300
refresh = 14400 (4 hours)
retry = 7200 (2 hours)
expire = 1209600 (14 days)
default TTL = 3600000 (41 days 16 hours)
```

```
it
primary name server = dns.nic.it
responsible mail addr = hostmaster.nic.it
serial = 2025032309
refresh = 10800 (3 hours)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 3600 (1 hour)
```

```
com
primary name server = a.gtld-servers.net
responsible mail addr = nstld.verisign-grs.com
serial = 1742727127
refresh = 1800 (30 mins)
retry = 900 (15 mins)
expire = 604800 (7 days)
default TTL = 900 (15 mins)
```

```
co.nz
primary name server = loopback.dns.net.nz
responsible mail addr = soa.nzrs.net.nz
serial = 2503231046
refresh = 900 (15 mins)
retry = 300 (5 mins)
expire = 604800 (7 days)
default TTL = 3600 (1 hour)
```

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

91

TTL: other examples

<pre>unipv.it primary name server = ipv36.unipv.it responsible mail addr = root.ipv36.unipv.it serial = 2025031902 refresh = 12000 (3 hours 20 mins) retry = 1200 (20 mins) expire = 1209600 (14 days) default TTL = 300 (5 mins)</pre>	<pre>polimi.it primary name server = ns.polimi.it responsible mail addr = root.ns.polimi.it serial = 2020006810 refresh = 21600 (6 hours) retry = 3600 (1 hour) expire = 604800 (7 days) default TTL = 86400 (1 day)</pre>
<pre>amazon.com primary name server = dns-external-master.amazon.com responsible mail addr = hostmaster.amazon.com serial = 2010193634 refresh = 180 (3 mins) retry = 60 (1 min) expire = 604800 (7 days) default TTL = 900 (15 mins)</pre>	<pre>twitch.tv primary name server = ns-219.awsdns-27.com responsible mail addr = awsdns-hostmaster.amazon.com serial = 1 refresh = 300 (5 mins) retry = 150 (2 mins 30 secs) expire = 604800 (7 days) default TTL = 60 (1 min)</pre>
<pre>google.it primary name server = ns1.google.com responsible mail addr = dns-admin.google.com serial = 739443398 refresh = 900 (15 mins) retry = 900 (15 mins) expire = 1800 (30 mins) default TTL = 60 (1 min)</pre>	

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

92

Reverse mapping

DNS lookup provides the IP address(es) associated with a given hostname (*forward mapping*)

It is also useful to obtain the hostname associated with a given IP address (*reverse mapping*)

This reverse lookup is implemented by a separate parallel tree (keyed by IP address) under reserved domains, i.e., **in-addr.arpa** for IPv4 and **ip6.arpa** for IPv6

The TLD is **arpa** (Address and Routing Parameter Area)

The nodes of a reverse mapping zone are administered by the owner of the block of IP addresses

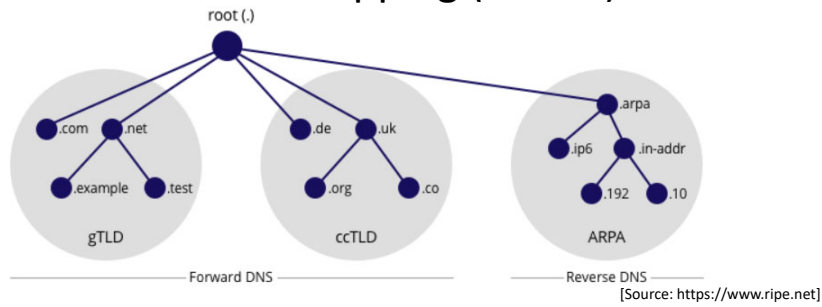
The reverse lookup process is based on the **PTR** resource record

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

93

Reverse mapping (cont'd)



What is the hostname associated with **192.58.128.30**?

Query: `30.128.58.192.in-addr.arpa IN PTR`

`30.128.58.192.in-addr.arpa. 3600 IN PTR j.root-servers.net.`

Answer: `j.root-servers.net`

Resource
Record

Used by web analytics tools (e.g., visitor identification) and for mail services (e.g., for fighting spam)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

Security risks

DNS is an *open distributed service* often seen as a *utility* whose design mainly focused on *availability, reliability* and *efficiency* of the service rather than on *security*

DNS is a *critical infrastructure* and any outage (e.g., due to security incidents) might have devastating effects

DNS is *robust*: during its long lifetime it survived many massive security attacks (with some disruptions of the service), e.g., “large” DDoS attack against Root Name Servers in 2002, huge DDoS attack of Mirai botnet against Dyn DNS provider in 2016

DDoS attack that disrupted internet was largest of its kind in history, experts say

Dyn, the victim of last week's denial of service attack, said it was orchestrated using a weapon called the Mirai botnet as the 'primary source of malicious attack'

Major cyber attack disrupts internet service across Europe and US

© M. C. Calzarossa 2025

Major DDoS attack on Dyn disrupts AWS, Twitter, Spotify and more

Well this is not good

October 21, 2016 By: Sebastian Moss Comment

Dyn DDoS Attack Highlights Vulnerability of IoT Devices

What can businesses do to guard against vulnerabilities as they bring more connected devices into their IT environments?

Enterprise Digital Infrastructure

95

Types of security risks

Two types of *security risks* affecting DNS service:

- Risks affecting *infrastructure* (i.e., Name Servers) are aimed at making service unavailable
- Risks affecting *data* (i.e., DNS database) are aimed at providing *fake* Resource Records and diverting traffic towards malicious servers

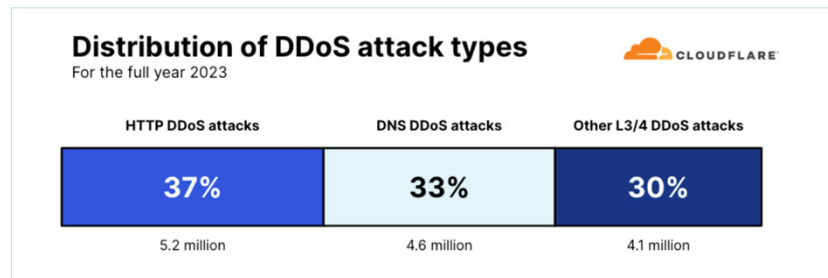
© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

96

DDoS attacks

DDoS attacks aim at exhausting Name Server resources and/or saturating the network



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

97

DDoS attack vectors

DDoS attacks leverage different techniques:

- Flooding the target Name Server (e.g., TCP SYN flood, UDP flood, NXDOMAIN attacks)
- Resolver and zone file vulnerabilities, such as unpatched software, Name Server misconfigurations (e.g., TsuNAME vulnerability with cyclic dependences)
- Amplification techniques (i.e., small queries with “big” response messages) to make DDoS attacks more aggressive and effective

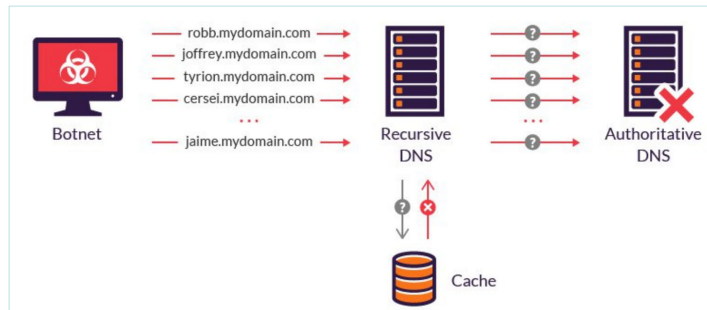
Reflection is typically used in the attacks, i.e., spoofed IP address to hide the actual source of the queries

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

98

NXDOMAIN attacks



Laundering attacks are based on a slow and methodical process very damaging in the long run

Subdomain prefixes are continuously randomized and not used more than once

Cache overflow of DNS resolver & overload of the second level (authoritative) NS

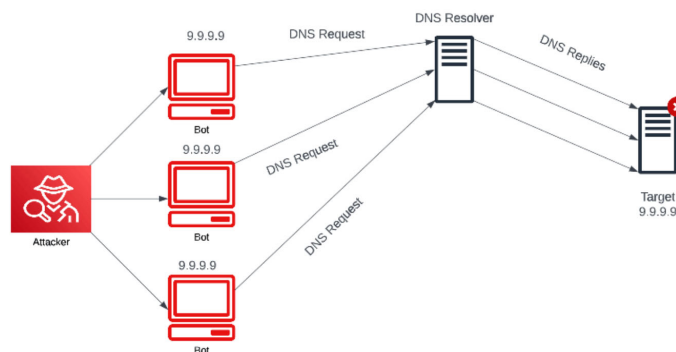
Difficult to mitigate because they originate from a legitimate Name Server

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

99

Reflection/amplification techniques



Bandwidth is saturated by big response messages (e.g., query of type **ANY**)

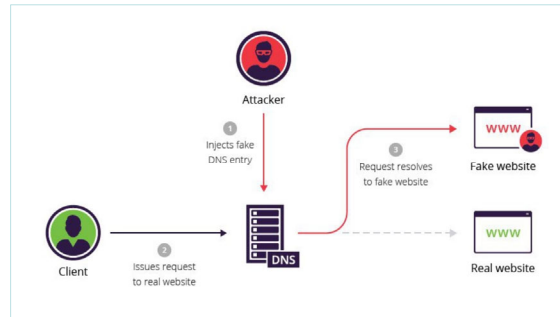
Both Name Servers appear to be under mutual attack

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

100

Security risks affecting RRs



There are multiple ways for intercepting and rerouting, or impersonating traffic on the Internet, so that the answer to a DNS query is *fake*

Malicious Name Servers deliver *fake* answers to DNS queries

Major impact on DNS caches (*cache poisoning*)

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

101

Security risks affecting RRs

Two main types of attacks against DNS database content

- *On path attacks*: the idea is to intercept the 16 bits query ID & source port number by eavesdropping the traffic and then race to generate a *fake* response message with the objective of diverting traffic towards malicious websites/services
- *Hijacking attacks*: the idea is to infect devices with malware with the aim of modifying their DNS settings and have them pointing to *rogue* Name Servers whose answers redirect users towards malicious websites/services

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

102

Recap

Serious security risks affect the Name Servers and the DNS database

DDoS attacks are employed to disrupt DNS services

On-path and hijacking attacks are employed to divert traffic towards services controlled by the attackers

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

103

DNS Security

DNS is a *critical infrastructure* not designed to be secure, so it is the ideal target for security attacks

Attackers can modify results of DNS queries with “incorrect” IP addresses

It is important to ensure the *integrity* of the DNS lookup results (i.e., correctness and authenticity of the IP addresses returned in the response messages)

Some security problems have been fixed

The Berkeley Internet Name Domain (BIND) forbids anything not related to the original request to be cached and rejects all the “out of zone” data

It is necessary to use mechanisms able to ensure that client resolvers can *trust* the answers to DNS queries

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

104

DNSSEC

One solution to fight counterfeit DNS answers is to add data origin *authentication* and *data integrity* assurance to the Domain Name System using digital signatures based on *public key* cryptography

DNSSEC refers to standardized DNS security extensions, that is, a hierarchical distributed trust system to validate the mappings of names to values [RFC 4033 “DNS Security Introduction and Requirements” published in 2005]

DNS Resource Records of a given domain are signed by the domain owner to allow the verification (i.e., proof of the identity of the Name Server that generates the answer) without having to trust the network or resolvers involved

DNS queries and responses are *not* cryptographically signed

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

105

Security extensions

DNSSEC extensions provide:

- a) origin authentication of DNS data allows a resolver to cryptographically verify that the received data comes from the zone where it believes the data originated
- b) data integrity protection allows the resolver to know that the data has not been modified in transit since it was originally signed by the zone owner with the zone private key
- c) authenticated denial of existence allows the resolver to validate that a certain domain name does not exist or to signal that a domain name exists, but does not have the specific RR type [RFC 7129 “Authenticated Denial of Existence in the DNS” published in 2014]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

106

DNSSEC implementation

Specific Resource Records (e.g., **DNSKEY**, **RRSIG**, **NSEC**) are used to implement DNSSEC

A public/private key pair is associated with each zone

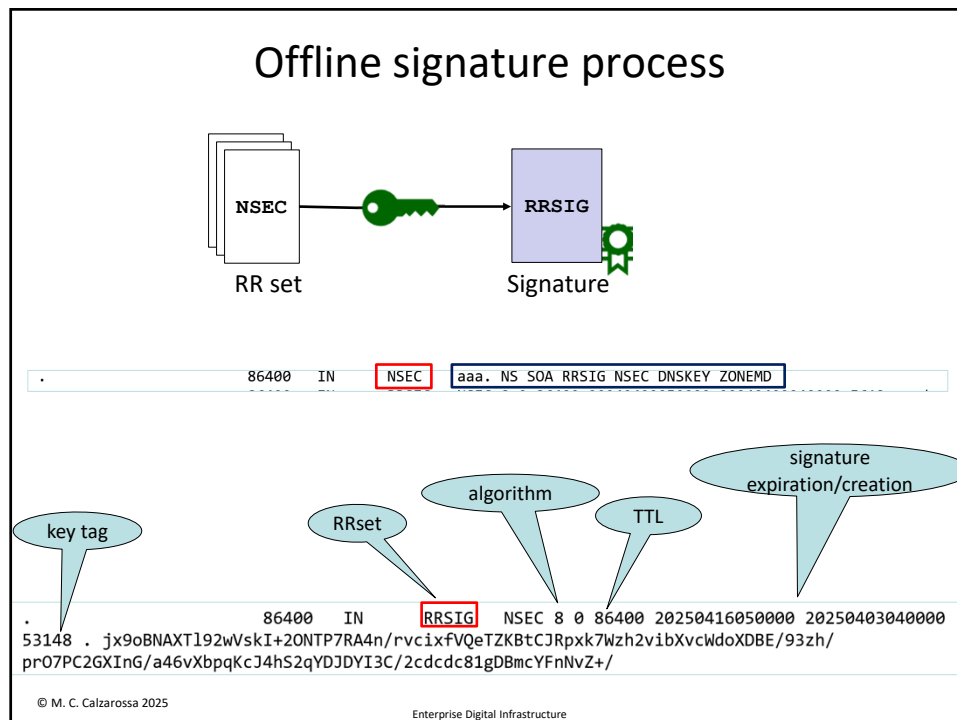
1. The zone public key (signed using the *parent zone* private key) is published in the DNS database [*offline* process]
2. The zone owner uses its private (secret) key to sign (i.e., encrypt) DNS *Resource Record sets* & stores digitally signed **RRSIG** records into its zone file [*offline* process]
3. Resolvers assess the authenticity of the DNS RRs using the zone public key
 - Digital signature of RR set valid: answer is legitimate and the resolver returns it to the client
 - Digital signature of RR set not valid: the resolver discards the data and returns an error to the client

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

107

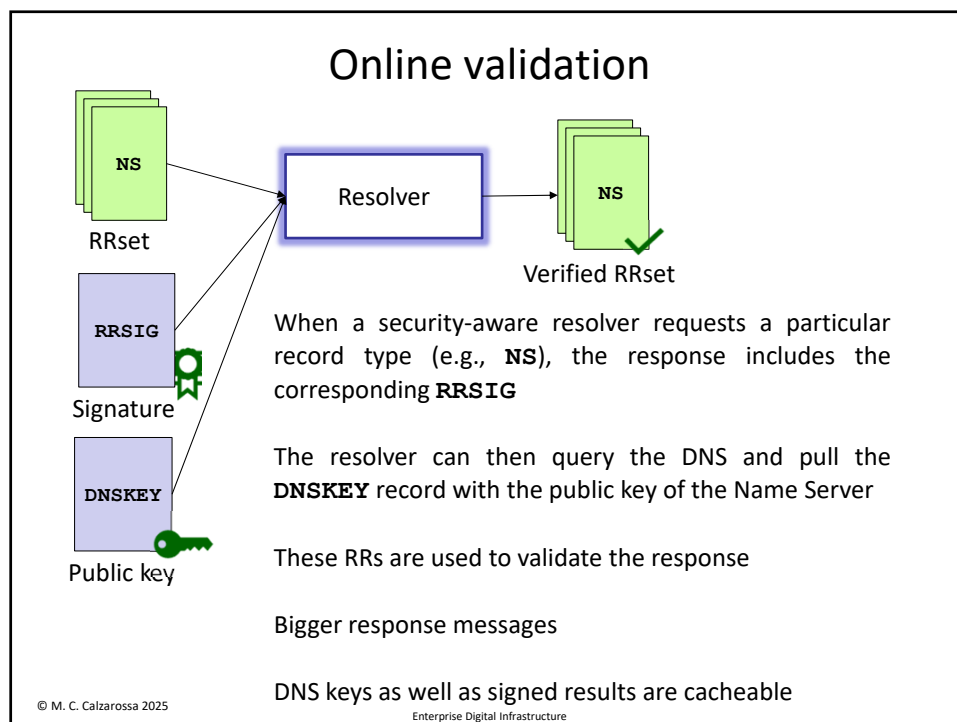
Offline signature process



© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

108



109

DNSSEC adoption

DNSSEC has not been adopted very extensively (about 35% worldwide DNSSEC validations, 27% for Italy)

Most TLDs are signed, only few second level domains

Many issues related to its deployment (e.g., complexity in the configuration, computational load, increased response time, security threats)

The partial deployment is causing problems (e.g., what to do with unsigned/unvalidated results)

Security depends on trust in Root Name Server's key and all other signing keys

DNSSEC does not ensure confidentiality

© M. C. Calzarossa 2025
Enterprise Digital Infrastructure

[Source: <https://stats.labs.apnic.net/dnssec>]

110

DNS Privacy with Speed

DNS traffic is still largely unencrypted, thus allowing both cybercriminals and ISPs to derive user profiles from plaintext DNS queries

The solutions proposed to ensure privacy in the lookup phases are based on the encryption of query and response messages

- DNS over TLS (DoT)
- DNS over HTTPS (DoH)
- DNS over QUIC (DoQ)

DoT and DoH are based on TCP transport protocol and on Transport Layer Security (TLS) encryption protocol

DoQ is based on the new QUIC transport protocol that combines transport and cryptographic features and ensures minimal latency

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

111

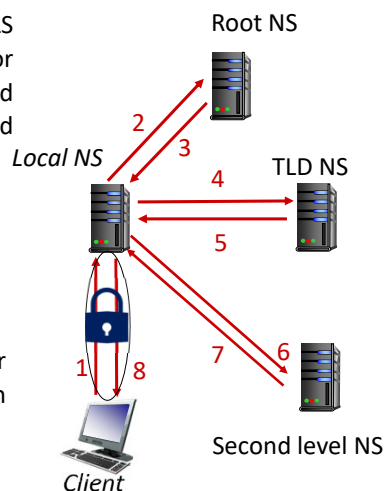
DNS over TLS (DoT)

The idea is to implement DNS lookups over TLS to ensure privacy [RFC 7858 “Specification for DNS over Transport Layer Security (TLS)” and RFC 8310 “Usage Profiles for DNS over TLS and DNS over DTLS” published in 2016 and 2018]

It complements DNSSEC and eliminates opportunities of eavesdropping and tampering of DNSSEC validated results on the “first leg”

Stub and recursive resolvers communicate over TLS-encrypted (*permanent*) TCP connections on port 853, thus ensuring channel integrity

Stub resolvers can be configured with *strict* privacy profiles or *opportunistic* privacy profiles to support incremental deployment

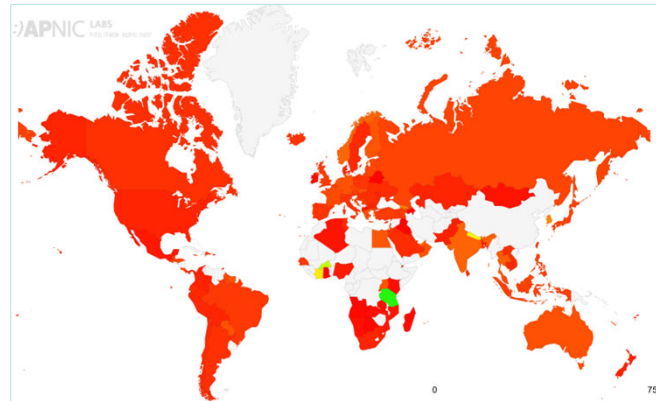


© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

112

DoT adoption



[\[https://stats.labs.apnic.net/edns?s=2\]](https://stats.labs.apnic.net/edns?s=2)

Only about 10.2% of the DNS queries are based on DoT

Private DNS under Android (i.e., **dns.google**) and public recursive European resolver (i.e., **dns0.eu**) implement DoT

© M. C. Calzarossa 2025

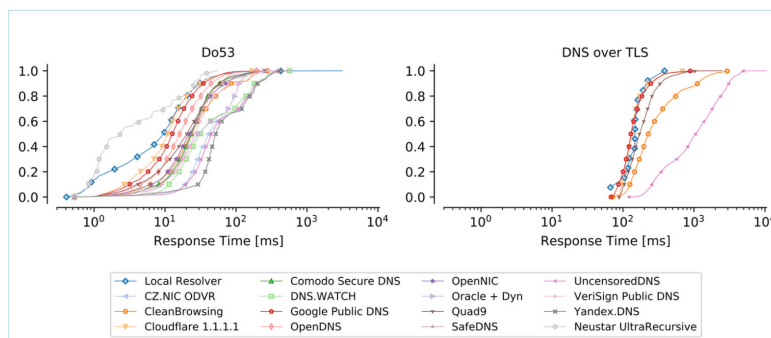
Enterprise Digital Infrastructure

113

DoT Performance

Measurements collected on the RIPE Atlas probes

Queries performed using DoT are slower by more than 100 ms compared to standard DNS queries (Do53) (e.g., overhead for the TCP and TLS handshakes)



[Source RIPE Atlas]

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

114

DNS over **https** (DoH)

Existing firewall configurations could block DNS queries over TLS

The idea is to implement DNS within **https** (or HTTP/2) to ensure privacy and confidentiality [RFC 8484 “DNS Queries over HTTPS (DoH)” published in Oct. 2018]

Each DNS query/response pair is mapped into an HTTP request/response pair

Queries are implemented using a **GET** method (i.e., content encoded in the URL) or a **POST** method (i.e., content included in the body as MIME type specified in **Content-type** header)

Standard port 443

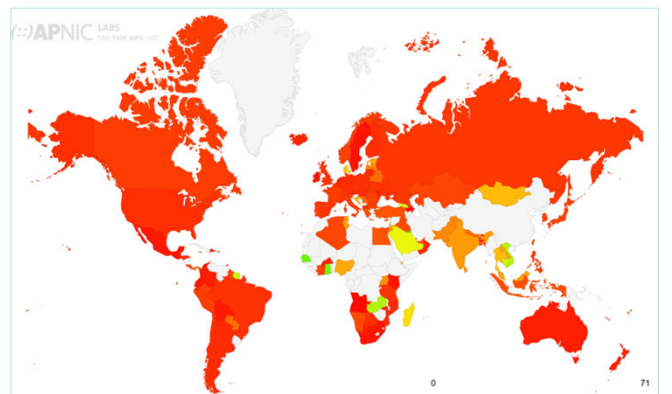
Responses are sent as HTTP response messages

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

115

DoH adoption



<https://stats.labs.apnic.net/edns?s=0>

Only about 13.3% of the DNS queries are based on DoH

Most browsers support DoH

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

116

Examples of query/response

Query: **pel.unipv.it IN A**

```
:method = POST
:scheme = https
:authority = dnsserver.example.net
:path = /dns-query
accept = application/dns-message
content-type = application/dns-
message
content-length = 33
```

pel.unipv.it IN A

Response: **193.204.34.1**
TTL= **3709** seconds

```
:status = 200
content-type = application/dns-
message
content-length = 61
cache-control = max-age=3709
```

193.204.34.1

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

117

DoH advantages & disadvantages

DoH provides *very good privacy* because it hides the services being accessed (unlike DoT)

DNS servers must be operated by *trusted* partners (they will see all queries)

Google, Cloudflare and Mozilla are pushing this protocol to promote privacy

DoH creates problems to ISPs and governments

Web surveillance (e.g., parental control, malware/website blocking) is often implemented by looking at DNS queries

In many countries ISPs are requested (by law) to store (for at least a year) the websites visited by citizens (e.g., monitoring of terrorism or other illegal activities)

Delivering content through CDNs becomes rather challenging

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

118

DNS over QUIC (DoQ)

DNS over QUIC [RFC 9250 “DNS over Dedicated QUIC Connections” published in May 2022]

Quick UDP Internet Connections (QUIC) is a UDP-based, stream-multiplexing, encrypted transport protocol

The encryption provided by QUIC has similar properties to those provided by TLS

DoQ has privacy properties similar to DoT and latency characteristics similar to classic DNS over UDP

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

119

DoT vs DoH vs DoQ

All these new protocols add desirable privacy and security features to DNS resolution

DoT is supported by operating systems (e.g., Linux, Android, MS Windows) and can be easily enabled, but its adoption rate is still quite low

DoH can be easily enabled inside browsers

Starting from Windows Server 2022, DNS clients support DoH

The impact of DoT or DoH on applications is generally not negligible

Geographic distance of recursive resolvers strongly affects the latency

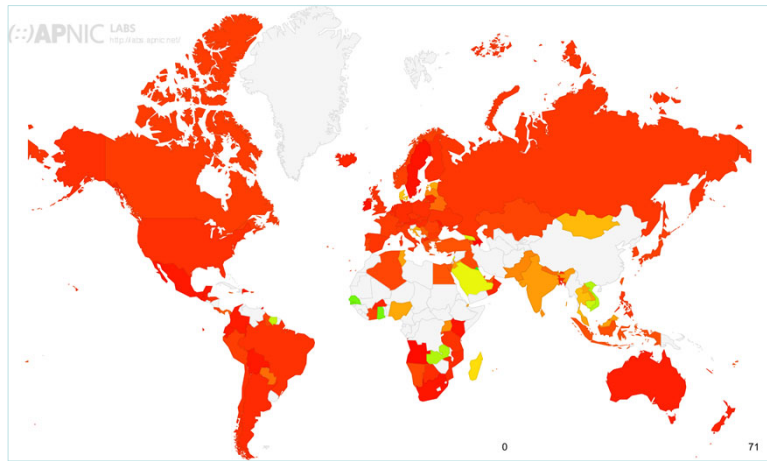
In general DoQ outperforms DoT and DoH in terms of DNS single query response time and its adoption by public DNS resolvers is slowly increasing

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure

120

Encrypted DNS



[\[https://stats.labs.apnic.net/edns?s=0\]](https://stats.labs.apnic.net/edns?s=0)

About 25.5% of the worldwide DNS queries are encrypted

Europe is at 17.5%, Africa and Asia at 30% each

© M. C. Calzarossa 2025

Enterprise Digital Infrastructure